INCH-POUND

MIL-W-19044E(SH)
18 September 1993
SUPERSEDING
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27 October 1986
(See 6.12)

MILITARY SPECIFICATION

WASHER-EXTRACTOR, LAUNDRY, 35 TO 200 POUND CAPACITY, MODIFIED COMMERCIAL (SHIPBOARD)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fully automatic combination washer-extractor machines for Navy surface shipboard use and are not suitable for use aboard submarines.

1.2 Classification. The washer-extractors covered by this specification are of the following types, classes, and styles, as specified (see 6.2):

Type I - Steam booster heater.
Type II - Electric booster heater.

Class 1 - Electro-mechanical automatic formula control unit.
Class 2 - Microprocessor automatic formula control unit.

Style A - Unsectionalized.
Style B - Sectionalized.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 03Q42, Department of the Navy, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3510

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. Standards. and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-F-320	- Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-H-2815	- Hose Assemblies, Rubber, Divers Breathing Air and Gas Supply
MIL-L-3153	- Laundry and Dry Cleaning Machinery and Equipment, Packaging of.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5	- Plates, Identification.
MIL-L-19140	- Lumber and Plywood, Fire-Retardant Treated.
MIL-H-24135	- l-lose, Synthetic Rubber, Wire Reinforced, and End Fittings, Reusable, for Flexible Hose Assemblies, General Specification for.
MIL-H-24135/10	- Hose, Synthetic Rubber, Wire Reinforced, and End Fittings, Reusable, for Flexible Hose Assemblies (Single or Double Wire Braid, Medium Pressure, Sizes -4 Through -48).
MIL-H-24136	- Hose, Synthetic Rubber, Synthetic Fiber Reinforced, and End Fittings, Reusable, for Flexible Hose Assemblies, General Specification for.
MIL-H-24136/3	- Hose, Synthetic Rubber, Synthetic Fiber Reinforced, and End Fittings, Reusable, for Flexible Hose Assemblies (Low Pressure Sizes -4 Through -32).

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services
FED-STD-313	- Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type 11 - Internally Excited).

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MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

MILITARY (Continued)
MIL-STD-740-1 - Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-1399, Section 300 - Interface Standard for Shipboard Systems Section 300 Electric Power, Alternating Current. (Metric)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents. drawings. and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

NAVAL WASHING FORMULAS SHIP STORE BULLETIN

(Application for copies should be addressed to Commander, Navy Resale and Services Support Office, Fort Wadsworth, Staten Island, NY 10305.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)
2000 - Gear Classification and Inspection Handbook - Tolerances and Measuring Methods for Unassembled Spur and Helical Gears (including Metric Requirements) (DoD adopted).
2001 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth (DoD adopted)

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standard Department 1500 King Street, Suite 201, Alexandria, VA 22314-2717).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 123 - Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products. (DoD adopted)
A 167 - Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip. (DoD adopted)
A 269 - Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service. (DoD adopted)
A 525 - Standard Specification for General Requirements for Steel Sheet, Zinc-Coated (Galvanized) by the Hot-Dip Process. (DoD adopted)
B 43 - Standard Specification for Seamless Red Brass Pipe, Standard Sizes. (DoD adopted)
B 61 - Standard Specification for Steam or Valve Bronze Castings. (DoD adopted)
B 62 - Standard Specification for Composition Bronze or Ounce Metal Castings. (DoD adopted)
B 75 - Standard Specification for Seamless Copper Tube. (DoD adopted)
B 209 - Standard Specification for Aluminum and Aluminum Alloy Sheet and Plate. (DoD adopted)
F 1166 - Standard Practice for Human Engineering Design for Marine Systems, Equipment and Facilities. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Boiler and Pressure Vessel Code
Section VIII - Rules for Construction of Pressure Vessels.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

THE INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS, INC. (IEEE)
45 - Recommended Practice for Electrical Installations on Shipboard.

(Application for copies should be addressed to The Institute of Electrical and Electronics Engineers, 445 Hoes Lane, P O. Box 1331, Piscataway, NJ 08855-1331.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)
MG 1 - Motors and Generators.

(Application for copies should be addressed to National Electrical Manufacturers Association, 2101L Street, NW, Washington, DC 20037.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
70 - National Electric Code.

(Application for copies should be addressed to the National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)
AS604 - Hose Assembly, Tetrafluoroethylene 400 Degrees F, 3000 psi Hydraulic, Heavyweight. (DoD adopted)
AS620 - High Temperature Hose Assembly, Convoluted Tetrafluoroethylene, for Aircraft.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.1.1 Material safety data sheet (MSDS). The contracting activity shall be provided a material safety data sheet at the same time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 6.9).

3.1.2 Human factors engineering. Human engineering criteria, principles, and practices, as specified in ASTM F 1166, shall be used in the configuration of the laundry washer-extractor where specified throughout this document. The washer-extractor shall be constructed so that access for operation, maintenance, and repair shall be analyzed in terms of the applicable requirements as specified in ASTM F 1166. Maintenance and operational activities shall permit performance by the 5th percentile female to the 95th percentile male as specified in ASTM F 1166. Access to each control, lubrication fitting, or any other component shall include space for the tools and test equipment required, as well as any protective clothing likely to be worn by the maintainer. Skill level and training requirements for operation, maintenance, and repair shall be kept at the lowest level consistent with other requirements herein. The analysis of operation, maintenance, and repair functions shall include and reflect considerations of safety.

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3.2 Materials. Materials shall be as specified in 3.2.1 through 3.2.7.

3.2.1 Structure. Frames, base, and strength members shall be constructed of q alleable iron, ductile iron, or steel hot-dipped galvanized in accordance with ASTM A 525 for corrosion protection. Gray cast iron or other brittle materials shall not be used for these parts. Except as otherwise specified herein, gray cast iron may be used for other components such as gear reducers and bearing housings, which are not strength members unless manufactured as an integral part of the frame and satisfactorily completes the vibration test procedures specified herein. Aluminum castings shall not be used.

3.2.2 Shell. The shell shall be made of corrosion-resisting steel in accordance with ASTM A 167, type 302, 304, or nickel-copper-aluminum alloy (minimum 60 percent nickel). Front of the shell shall be of corrosion-resisting steel metal or ferrous metal with corrosion resisting metal liners on the side exposed to the washing solution. The shell shall be provided with nickel-copper-aluminum alloy (minimum 60 percent nickel) or corrosion-resisting steel door frame.

3.2 3 Cylinder The cylinder shall be made of corrosion-resisting steel in accordance with ASTM A 167, type 302, 304, or nickel-copper aluminum alloy (minimum 60 percent nickel). The driving trunnion and spider shall be of the same material as the cylinder, or of ferrous metal protected by hot-dip galvanizing in accordance with ASTM A 525, or corrosion-resisting coating.

3.2.4 Metal joints Rivets, bolts, machine screws, nuts, and washers shall be of steel, except where brass or corrosion-resisting metal is fastened, in which case they shall be of brass or corrosion-resisting metal. Where dissimilar metals are fastened, rivets, bolts, machine screws, nuts, and washers shall be galvanically compatible with materials being joined. Electrodes or welding rods used for welding shall deposit material similar to the base metal. Nuts subject to shock or vibration shall be locked. Welded steel parts, required to be galvanized, shall be galvanized after completion of welding in accordance with ASTM A 123. Drill point screws are prohibited. Spacing of fasteners shall provide strength, tightness, and rattle-free performance. Fasteners of parts through which servicing access is required shall be provided for convenient access.

3.2.5 Pipes. fittings. valves. and connections.

3.2.5.1 Pipes. Pipes shall be made of corrosion-resisting steel, grade TP 304, in accordance with ASTM A 269 or of brass in accordance with ASTM B 43 or of copper in accordance with ASTM B 75.

3.2.5.2 Flexible steam hose assemblies. Flexible steam hose assemblies shall be extruded or convoluted tetrafluoroethylene hose with stainless steel wire braid and end fittings in accordance with SAE AS604 or AS620.

3.2.5.3 Flexible hot and cold water hose assemblies. Flexible hot and cold supply water hose assemblies shall be in accordance with MIL-H-24135 and MIL-H-24-135/10 or MIL-H-24136 and MIL-H-24136/3. Hoses for automatic supply injection assembly to be of a 250 pounds per square inch working pressure and 212 degree Fahrenheit shall be in accordance with MIL-H-2815 RMA (class A).

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3.2.5.4 Filtting. Fittings shall be made of bronze In accordance with ASTM B 61 or ASTM B 62 or of stainless steel.

3.2.5.5 Valves. Valves shall be made of cast bronze in accordance with ASTM B 61.

3.2.6 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless othewise specifically specified.

3.2.7 Restricted materials The machine shall be free from asbestos, wood, or cadmium parts or coatings in any material form.

3.3 Configuration. The complete machine and its components shall be new and shall be the current standard product of the manufacturer, modified as necessary to meet the requirements of this specification.

3.3.1 General construction. The machines shall be the front loading, fully automatic, shell and cylinder (either multi-pocket, or open pocket for washer-extractors 60-pound capacity and less) type with a rotating action about the horizontal axis. Axial rotation of cylinders shall incorporate reversing action during appropriate parts of the production cycle. The machines shall operate efficiently with soaps or detergents. Each machine shall be fitted with features as specified herein.

3.3.1.1 Automatic controller. Each machine shall be furnished with an automatic formula control unit for controlling the machine through any complete cycle of wash, multiple rinse, drain, and intermediate and final extraction phases Up to 60 minutes in duration. The automatic formula control unit shall be the electro-mechanical type on class 1 washer-extractors and solid state, microprocessor type on class 2 washer-extractors.

3.3.1.2 Supply dispensers.

3.3.1.2.1 Automatic supply. Each machine shall have an automatic supply dispenser for adding supplies as directed by the automatic formula control unit. The dispenser shall be corrosion resistant and shall have not less than four compartments which may be filled manually at the beginning of the wash cycle. Supply dispenser compartments shall be of a water flushing type Automatic supply dispensers shall not allow soap or detergent to clog or cake. For supply control requirements, see 3.4.8.2. Tilt cans, dump cans, or supply cups shall not be used.

3.3.1.2.2 Manual supply. A separate machine opening shall be provided for manual addition of supplies. The opening shall be designed to protect against water splashing or overflowing during normal operation or in the event the machine should overfill.

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3.3.1.3 Booster heater. Each machine shall be equipped with a booster heater unit. One of the following two types of booster heaters shall be provided.

(a) Indirect steam heating for type I machines.
(b) Electric heating for type II machines.

3.3.1.4 Door interlock. The shell door on each machine shall be fitted with an electric interlock that will prevent the door from being opened when the machine is operating.

3.3.1.5 Automatic values. Drain valve, hot and cold fresh water inlet valves, and steam inlet valve (on type I washer-extractors) shall be automatic. Valves shall be readily accessible for maintenance (see 3.1.1).

3.3.1.6 Vibration switch. The machine shall have an adjustable vibration limit cut-out device.

3.3 1.7 Capacity. size . and weight. The capacity, size, and weight of the machine shall be in accor- dance with table I, as specified (see 6.2).

TABLE I. Capacity, size and weight.

Normal capacity per load up to this dry weight (pounds)	Overall dimensions (inches maximum)			Maximum weight (pounds	Cylinder volume (minimum ft^3)
	Length (across front)	Depth (front to rear	Height		
35	55	60	72	1700	6.0
60	60	66	81	2500	12.0
100	66	69	86	4500	18.5
135	80	70	86	5200	25.0
200	82	79	86	5600	36.0

3.3.2 Water consumption. Machine shall be not greater than the following water consumption when run through the standard Navy wash formula (see table IX) for all types and style machines;

Capacity per load up to and including this dry weight (pounds)	Maximum water consumption (gallons per cycle)
35	100
60	175
100	200
135	260
200	480

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3.3.3 Machine mounting. Washer-extractor mounting to deck shall be provided by a minimum of four mounting bolt connections not less than 1/2-inch nominal diameter. No other attachment points are permitted except for water lines, air lines, steam lines, electrical cabling, and drain lines. Base structure of mounting pads shall be secured to an area of the structure to ensure adherence of the structure and its attachments under ship motion and vibration. Structures penetrated by the mounting bolts shall be made of steel and shall be constructed so as be not greater than the yield point of its material when subjected to the force (washer-extractor weight times G's) specified in table II about the center of gravity of the washer-extractor.

TABLE II. Acceleration of gravity (G) about the center of gravity of washer-exractor to determine mounting.

Normal capacity up to this dry weight (pounds)	Maximum weight (pounds)	G's vertical	G's horizontal
60	2500	52	21
100	4500	44	18
200	5600	41	16

3.3.4 Inclined operation. The machine with automatic formula control unit installed shall function in accordance with the requirements specified herein without reduction in performance or spilling greater than a total of 1 gallon of water when operated at an angle of 12 degrees to each side of two vertical planes at right angles to each other as specified in 4.4.1. Spillage, if any, shall not damage electrical parts. The booster heater shall remain submerged in water while the machine is inclined to prevent burning out the element.

3.3.5 Vibration. Washer-extractor, with automatic formula control unit installed, shall meet the applicable requirements of MIL-STD-167-1 from 4 to 33 hertz (Hz) for type I vibration (see 4.4.2).

3.3.6 Drives.

3.3.6.1 Cylinder drive. The cylinder shall be driven by an electric motor or motors through V-belts or a combination of gearing and V-belts to provide proper speed for the washing, extracting, and draining (distribution) cycles. The cylinder shall reverse rotation a minimum of three times per 95 seconds during the washing cycle. Means shall be provided for inching the cylinder door to the proper loading position on multi-pocket machines (see 3.4.7.4.5). Means shall be provided for adjusting tension of the V-belts.

3.3.6.2 Gears. Gears shall be in accordance with or exceed all provisions of AGMA 2000-A88 and AGMA 2001-B88. The gears shall be of proper design to transmit full rated horsepower throughout the speed ranges without failure for the expected service life of the machine.

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3.3.6.3 Bearings. Bearings shall be self-aligning, readily replaceable and properly sized. Seals shall be provided to make bearings waterproof. Cylinder trunion bearings shall be self-aligning, shielded, antifriction bearing type.

3.3.6.4 V-belts. V-belt drives shall be selected in accordance with the belt manufacturer's recommendations for load ratings, including applicable safety factors, to drive the machine under the most severe operating conditions. The V-belts shall perform for a minimum service life of 5,000 hours of operation under normal operation and with proper maintenance.

3.3.7 Extraction efficiency. Water retention shall be not greater than 55 percent after 5 minutes of extraction when tested as specified in 4.4.3

3.3.8 Airborne noise. The washer-extractor shall be not greater than the octave band sound levels specified in table III when tested under full load capacity conditions in both the wash and extraction modes of operation (see 4.4.4).

TABLE III Acceptance octave band sound pressure levels (in decibels (db) re 20 Pascals (Pa)).

Equipment grade	Octave band center frequency (in Hz)								
	31.5	63	125	250	500	1000	2000	4000	8000
D	91	88	85	82	79	76	73	70	67

3.3.9 Finish and color. The washer-extractor shall be finished according to the contractor's standard commercial practice.

3.3.10 Sectionalization. Style b washer-extractor shall be constructed to be disassembled, transported, and reassembled within the ship without hot work, such as welding and brazing. Major subassemblies that bolt together shall be preferred. Limiting dimensions for each major subassembly shall be 50 inches high, by 25 inches wide, by 36 inches deep. Subassemblies shall be match-marked to assist in field reassembly once in the designated space. The manufacturer shall provide touch-up paint in the event of chipped or scratched paint during breakdown and assembly of washer-extractor A replacement set of bolts, nuts, and other associated fasteners or hardware shall be provided in the case of lost or damaged items during breakdown and assembly.

3.3.11 Interchangeability. All parts of the end item having the same manufacturers part number shall be manufactured to definite standards and tolerance that will provide for the interchangeability of respective replacement parts between end items without modification of the part or the machine.

3.3.12 Lubrication. Means shall be provided for easy accessibility to lubricate mechanisms requiring periodic manual lubrication.

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3.3.13 Threads. All threaded parts used on the machine and its related attachments and accessories shall be inaccordance with FED-STD-H28 and the applcable "Detailed Standard" section referenced therein.

3.4 Detailed requirements.

3.4.1 Shell. The shell shall be watertight and rigid to prevent permanent distortion or objectionable vibration under the normal operation on board ship. The shell shall be provided with connections for piping, and shall be reinforced where necessary to properly support attachments. The front of the shell shall be removable to permit replacement of the cylinder and the shell seams shall not pass through the drain valve opening.

3.4.1.1 Shell openings. The shell shall have openings for a door, cylinder shaft, four-compartment automatic and manual supply dispensers, water and either steam piping or electric element wiring, drain outlet, overflow and vent outlet, and water level and temperature Indicator. The manual supply dispenser opening shall be provided for manual addition of supplies prior to the beginning of the wash cycle. The entry shall preclude backflow due to either splashing within the machine or an abnormal overflow condition. The shell shall be provided with a door mounted on corrosion resisting heavy duty hinges, complete with latching means to ensure watertightness when closed. When the door is fitted with a window, the tempered glass used shall be of a quality and thickness to prevent damage under normal operation. The glass shall be sealed and be readily replaceable. For door interlock see 3.3.1.4.

3.4.2 Cylinder. The cylinder shall be rigid to prevent distortion or failure from stresses created during the extraction cycle at rated capacity. Perforations in the cylinder shall be sized and spaced for efficient washing, rinsing, and extraction. Perforations shall be free from burrs, sharp edges, or imperfections that will adversely affect clothing If the cylinder is of the three-pocket type, it shall be provided with a hinged door as part of the front head and means to prevent laundry from falling between the cylinder and shell during loading and unloading. The door shall provide not less than 150 square inches (in^2) area for each 50 pounds of rated machine capacity.

3.4.2.1 Cylinder shaft. The cylinder shaft openings shall be sized according to shaft size and fittings. Shaft openings shall be sealed to prevent water spillage during machine operation.

3.4.3 Brake. The machine shall be fitted with a electric or pneumatic power operated brake for rapid deceleration and stopping of cylinder. The brake shall stop the rotation of a loaded cylinder during any part of the extraction cycle within 60 seconds. The temperature rise of the friction surface of the brake lining shall be not greater than 300 degrees Fahrenheit (°F) above the initial temperature of the brake lining as a result of three stops from peak speeds at 2-minute intervals during the extraction cycle.

3.4.4 Discharge outlet. A normally open, automatically controlled drain valve shall be fitted at the lowest point of the shell. The valve shall be operated electrically or pneumatically. The discharge pipe shall be not less than 2-1/2 inches in diameter for the 100-pound and under machines and 4 inches in

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diameter for the over 100-pound machines. The drain valve shall be made of commercial bronze or corrosion resisting stainless steel in accordance with ASTM A 167, series 300 or nickel-copper-aluminum alloy trim in accordance with ASTM B 209, and fitted with a replaceable resilient seat

3.4.5 Piping.

3.4.5 1 Piping connections. The water inlets shall be connected to the machine a q inimum of 3 inches above the maximum high water level. Piping valves and fittings shall have a working pressure of 100 pounds per square inch (psi) A vacuum breaker backflow preventer shall be installed in the water inlet piping system.

3.4.5.2 Inlet valves. Two separate electrically or pneumatically controlled inlet valves for hot and cold fresh water shall be provided. The valves shall be not less than a threaded 1-1/4 inches nominal pipe size (nps) for the 100-pound and under machines, and not less than a threaded 1-1/2 inches nps for the over 100-pound machines. Valves shall be accessible for maintenance and replacement. The machine shall be fitted with a threaded 1-1/2 inch nps salt water inlet connection.

3.4.5.3 Overflow and vent connections. The machine shall be fitted with combination overflow and vent connections located so that the overflow may be piped to the drain line when the machine is installed. Connections shall be of a size that will prevent the water from splashing or spilling from any part of the machine, other than the overflow connection, when both hot and cold water valves are fully opened and being supplied at 40 psi inlet pressure.

3.4.5.4 Steam connection. The type I machine shall be provided with a steam connection not less than a threaded 1/2 inch nps The steam heat exchanger shall be a closed loop type; that is, steam shall not come in contact with cylinder wash or rinse water. Steam systems shall be hydrostatically tested as specified in 4.7.6. During testing, no leaks shall be permitted in the steam system. Pipes, fittings, coils, valves, and chambers using steam shall be suitable for 100 psi saturated steam working pressure.

3.4.6 Water leve l and temperature indicators. The water level shall be visible from the front of the machine and shall be calibrated in inches. A thermometer, either separate or incorporated in the temperature controller, shall be provided to indicate temperature of water in the shell with a range of not less than 60 to 220°F.

3.4.7 Electrical requirements. The electrical equipment shall conform to the recommended practices for electrical installation on board ship in accordance with IEEE 45. Electrical equipment shall operate on a type I electrical system in accordance with MIL-STD-1399, section 300. The control panel shall be dripproof in accordance with MIL-STD-108. Pushbuttons and switches shall be arranged for safe and convenient operation to ensure operator hands are safely outside rotating cylinder area.

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3.4.7.1 Electric motors. Electric motors shall be in accordance with NEMA MG 1 and shall conform to the following characteristics:

Rating	-	Continuous.
Insulation	-	Class B or F.
Voltage	-	440 volts alternating current (Vat), 3-phase, 60 Hz.
Enclosure	-	Dripproof or better.
Speed		Wash motor: constant, reversing a minimum of 3 times per 95 seconds.
		Extract motor: multispeed.
Protection	-	Thermally protected.

3.4.7.1.1 Wire. Motor shall be wound with wire sizes conforming to American Wire Gauge (AWG). Half size wire shall not be used.

3.4.7.1.2 Bearing. Motors shall be equipped with ball bearings, except motors of l/2-horsepower or less when used for horizontal applications may use sleeve bearings. Motor bearings shall be permanently lubricated type or shall have adequate and accessible means for lubrication

3.4.7.1.3 Thermal protection. Each motor 1/8-horsepower rating or larger shall be provided with manually reset thermal overload protection incorporated as a part of the motor or installed in the controller.

3.4.7.2 Controllers. Motor controllers shall be of the magnetic type with overload protective relays. Contractors shall switch all three lines of 3-phase circuits and both sides of single-phase power circuits. Motor controllers shall provide thermal overload protection for motors 1/8-horsepower and larger. Control circuits shall operate at 240 volts or less. Reduced voltage control circuits shall be protected by cartridge fuses sized not greater than 200 percent of the transformer kilovoltampere (kVA) rating or 100 percent of control wire amperage rating, whichever is smaller. Start, stop, and jog pushbuttons and interlocks shall be commercially dripproof. Tumbler type switches are not permitted. Electrical components mounted outside the control panel shall be provided with dripproof enclosures in accordance with MIL-STD-108 or better. Solenoid valves shall be provided with epoxy (high temperature) encapsulated coils. Control boxes and electrical or air operated control valves shall be accessible from the sides (front or back) of the machine to the maximum extent practicable.

3.4.7.3 Wiring methods and practices. Wiring between different devices mounted on the machine routed outside of an enclosure shall be in rigid metal conduit or liquid tight flexible metal conduit not less than 3/8 inch electrical trade size. Fittings and junction box connections for conduit entrance shall be either threaded bushings or watertight, neoprene bushed, strain relief type. Drainage holes shall be provided at the low point of all conduit runs or junctions. Wiring shall be with moisture resistant thermoplastic insulated copper conductors rated for an operating temperature of 176°F. Conductors shall be stranded. Wiring shall be color coded and marked every 12 inches with terminal numbers. Terminal blocks shall be plainly marked to correspond with the markings on the electrical diagram. Wires run in conduit or in harness assemblies shall be marked to correspond with the electrical diagram. Solder lugs, soldered joints, and wire-nut type connections are not permitted. Electrical connections shall be solderless ring connectors. The ends of each wire shall be connected to terminals

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on the part or to the terminal boards by means of a pressure or solderless lug terminal. Not greater than three connections shall be made at each terminal. Conduit, fittings, and wire shall conform to and be installed in accordance with NFPA 70. Equipment shall be used on an ungrounded electrical system and no grounds shall be permitted on the machine wiring. Leads shall be extended from motors and control devices to a connection box and properly Identified for connections to wiring beyond the unit. Electric cable length shall be kept to a minimum with sufficient use of wire wraps and clamps to prevent wire insulation chaffing. The electrical controls shall be provided with a hinged access door with hardware to tightly secure it closed. (Key latches and magnetic latches shall not be used.) Maintenance tasks shall be easily and safely completed by the 5th percentile female to the 95th percentile male.

3.4.7.3.1 Electromagnetic Compatibility (EMC). The equipment shall be tested to the applicable electromagnetic interference (EMI) requirements of MIL-STD-461 class A4, with the following modifications:

(a) The equipment shall be tested for the collection of CE01, CE03, and RE02 electromagnetic interference data.

3.4.7.4 Interlock and safety devices.

3.4.7.4.1 Wash-extract interlock. The driving arrangement shall permit changing from washing to extracting cycle and vice versa with interlocking features to ensure smooth transfer.

3.4.7.4.2 Functional modes for interlocks and safety devices. Safety and interlocking devices shall function for both clockwise and counterclockwise rotation of the cylinder and when ship is rolling or listing 12 degrees from normal position.

3.4.7.4.3 Brake interlock. The machine shall be fitted with a power operated brake as specified in 3.4.3. A brake interlock switch shall be provided to prevent extraction with brake applied.

3.4.7.4.4 Door interlock. The shell door shall be interlocked to prevent rotation, except for inching, when the shell door is open and to prevent the shell door from being opened until the cylinder is at rest.

3.4.7.4.5 JOG buttons. Jog buttons shall be provided on multi-pocket washer-extractors. The JOG buttons shall require the use of both hands of the operator for inching the cylinder. Depressing both JOG buttons shall release the brake to permit cylinder and shell door alignment JOG buttons shall be located near the cylinder door area.

3.4.7.4.6 Guards. Exposed belts, shafts, pulleys, and other moving parts shall be fully enclosed by q etal guards. Gears shall be enclosed in a gear housing. Metal guards shall be firmly supported but readily removable for maintenance.

3.4.7.4.7 Vibration switch. The machine shall be furnished with an adjust-able vibration limiting device to disconnect service should excessive, equipment damaging vibration occur. A red indicator light shall be provided to identify the activation of the vibration switch.

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3.4.7.4.8 Drain interlock. The machine drain valve shall be electrically or pneumatically interlocked to prevent the machine from entering or con- tiuing in the extract mode when drain valve is closed.

3.4.7.4.9 Emergency stop button. Emergency stop button shall be located on the front of the machine within an accessible area. Emergency stop button shall be bright red with the words "Emergency Stop" printed with white lettering on the face, with a minimum height of 3/8 inch. Minimum dimensions on the emer- gency stop button shall be 1 inch or a diameter of 1-1/4 inch. Button shall bring machine to an immediate stop when activated. Machine shall be reset with a reset button. The activation of the emergency stop button shall be identified by a red indicator light.

3.4.8 Automatic formula control and supply dispensing. Each machine shall be provided with an automatic formula control unit for directing the machine through a complete operating cycle (washing, rinsing, extracting) as programmed, and an automatic supply dispenser for adding supplies as directed by the formula control unit.

3.4.8.1 Automatic formula control unit. The unit shall be of the replace- able plastic chart type on class 1 washer-extractors and a microprocessor con- troller on class 2 washer-extractors. The unit shall automatically control fills, drains, temperatures, rinses, extracts, supply injections, and shall signal, both audibly and visually, at the end of the cycle or at other programmed times.

3.4.8.1.1 Control phases. The control unit shall automatically control all phases of clutching and braking, and all motors.

3.4.8.1.2 Power input and output controller connector. For bulkhead mounted controllers (see 3,4.8.1.11 a single or separate quick-locking-type connector shall be used to connect the power inputs and outputs to the automatic formula controller. The configuration of the connector or connectors shall prevent incorrect wire attachment from the washer-extractor to the automatic formula controller and shall prevent accidental disconnection.

3.4.8.1.3 Functional control. The control shall automatically control the water level for three separate independent levels which shall be readily readjustable in the field. Control shall automatically add water when needed to compensate for the water absorbed by the load. Automatic timing of each operation shall not proceed unless the proper water level and water temperature exists. Water temperature control shall be achieved by thermostatically regulat- ing the hot and cold fresh water inlet valves during filling phase, and after the level has been attained, by automatically thermostatically regulating steam to the machine for type I machines, and by automatically thermostatically energizing or de-energizing the immersion heater on type II machines. Means shall be provided to allow the operator to select and set and infinitely adjust not less than two controlled water temperatures from 60 to 220°F. Indicators shall provide analog or digital display of the set point bath temperature, The temperature selectors shall be electrically interconnected to the automatic formula control water temperatures for any bath or baths 60 to 220°F as programmed. Booster heater shall be automatically activated when tub water level-has been obtained . Drain shall not be activated until booster heater is deactivated and vice versa.

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3.4.8.1.4 Pilot light. A pilot light shall be provided to indicate when the automatic timing is proceeding on class 1 washer-extractors. Class 2 washer-extractors shall have a digital LED or LCD timer display Indicating time left within the formula.

3.4.8.1.5 Fresh or salt water selection. Means shall be provided on the automatic control unit to manually open and close the hot and cold water valves or allow the operator to manually fill the machine with salt water.

3.4.8.1.6 Audible and visible signals. A control shall be furnished to provide audible and visual means to signal the operator when programmed , and at the end of the formula. The signal shall persist until canceled by the operator. Automatic timing shall not proceed until the signal is canceled. With the exception of when the machine is in the extraction phase, the machine shall stop whenever the signal operates and shall restart when the signal is canceled.

3.4.8.1.7 Simultaneous energizing. The control and machine shall be so wired that simultaneous energizing of greater than one phase of formula at a time will not occur.

3.4.8 1.8 Drain selection. A means shall be provided on the automatic control unit to manually open and close the drain valve and shall be interlocked with heating element to de-energize element when drain is opened.

3.4.8.1.9 Booster heater selection. A means shall be provided on the automatic controller to manually activate the booster heater.

3.4.8.1.10 Microprocessor controller. The microprocessor formula controller unit on class 2 washer-extractors shall have an operational mode for storing a minimum of four independent laundry wash formulas. The four formulas shall have a minimum of 25 separate steps (that is, break suds, rinse, extract, drain, and finish). Controller shall be provided with a means to start, stop, or reset the existing selected formula. Current naval washing formulas may be obtained from the “Ship Store Bulletin”.

3.4.8.1.10.1 Nonprogrammable controllers. Nonprogrammable controllers shall have formulas contained on a single, replaceable etched electronic chip. Replaceable chips shall snap in and out of microprocessor circuitry without any type of soldering, rewiring, or special tools.

3.4.8.1.10.2 Programmable controllers. Programmable controllers shall retain formulas in static random access memory, incorporating a back-up power source to maintain the memory in case power is lost to the washer-extractor. The back-up power source shall be recharged when the power is restored to the washer-extractor. A locking mechanism shall be provided to prevent accidental reprogramming or tampering by untrained and unauthorized personnel. Programmability shall incorporate the ease of step selection within an existing programmed formula or a formula undergoing programming. The selection of limits (for example, water temperature, step time, and so forth) for required steps shall be programmed. If the step does not have the proper limit or limits specified, or the limits specified are unreasonable (for example. 32°F bath temperature) the step shall indicate a programming error and further programming may not commence until the error is alleviated. An end step shall indicate to microprocessor to assign

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formula steps into memory. The circuitry for the microprocessor shall use bread boards which can be removed and replaced using a plug-type receptacle. The mounting of the bread boards within the microprocessor shall prevent electrical disconnection or shorting under adverse washer-extractor out-of-balance forces and vibrational forces. Operation and programmability (when required) of the microprocessor shall be at an 8th grade level.

3.4.8.1.10.3 Displays, buttons, and switches on controller. The displays, buttons, switches, and so forth, on the controller shall be clearly labeled in plain english and shall not be confused with other labeled displays, buttons, and switches. As a minimum, the following displays, buttons, and switches shall be provided:

Power	- A display light and switch/button shall be provialed to turn on and off the electrical power to the microprocessor.
Formula selector	- Allows the operator to select a wash formula from memory An indicator shall be provided to identify the wash formula under selection and operation.
Drain	- A display shall be provided to indicate the drain cycles contained in a programmed wash formula. Programmable microprocessors shall use the display to indicate the programming of the drain step within the wash formula.
Timer	- A digital display shall be provided to indicate the total time remaining in the selected wash formula. The timer shall descend when the wash formula is in operation and shall have a maximum time increment of 1 minute. Programmable microprocessors shall also use the digital timer display to program the time for each required step within the wash formula.
Supplies (1,2,3,4)	- Displays shall be provided to indicate the chemical supply compartments being dispensed during the wash formula. Programmable microprocessors shall also use the chemical supply compartment displays to program the chemical supply compartments needed for each required bath step within the wash formula.
Water levels (low, medium, and high)	- Displays shall be provided to indicate the water fill height selected during the wash formula operation. Programmable microprocessors shall also use the fill height indicators to program the water fill height needed for each required bath step within the wash formula.
Bath temperature	- A digital display shall be provided to indicate the bath temperature within the wash cylinder if an analog/digital display is not provided elsewhere on the washer-extractor (see 3.4.6). The digital display on the microprocessor shall have a minimum resolution of 1°F.
Set point temperature	- A digital display shall be provided to indicate the bath temperature set point. Programmable

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	microprocessors shall also use the digital display to program the water temperature for each required bath step within the wash formula. A digital display shall have a minimum resolution of 1°F.
Signal	- A push button shall be provided to cancel the signal. Programmable microprocessors shall use a signal display to program the signal operation within a wash formula.
Start/resume	- A push button shall be provided to start a selected wash formula or a wash formula which has been stopped.
stop	- A push button shall be provided to stop a selected wash formula in process.
Reset	- A push button shall be provided to reset a stopped wash formula to it's beginning to be restarted or to allow selection of another wash formula.
Extract	- A display shall be provided to indicate the extract cycles of a wash formula. Programmable microprocessors shall use the display to program the extract step process within a wash formula.

3.4.8.1.11 Mounting controller. The automatic formula control unit may be mounted remotely or directly to the washer-extractor. When the control unit is directly mounted it shall permit easy viewing and accessibility and shall not interfere with the operation or maintenance of the washer-extractor. The control unit's mounting shall inhibit the transfer of vibration and provide a dripproof seal for the controller. The controller shall provide a minimum of 8 feet of cable length for remote mounting.

3.4.8.1.12 Automatic controller test Performance. Control units shall be tested for the tests (see 4.4.6) and performance requirements as follows:

Test	Performance requirement
Low temperature	To determine if controller can be stored and operated under the conditions specified in 4.4.6 without experiencing physical damage or deterioration in performance.
High temperature	To determine if controller can be stored and operated under the conditions specified in 4.4.6 without experiencing physical damage or deterioration in performance.
Humidity	To determine controller's ability to perform in hot and humid atmosphere (see 4.4.6) without physical damage or deterioration in performance.

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Enclosure	Failure of control unit to operate or accumulation of water within the enclosure shall be cause for rejection (see 4.4.6).
Steady state voltage and frequency	Control unit shall operate satisfactorily after each test condition (see 4.4.6).
Transient voltage and frequency	Control unit shall return to normal operation without deterioration in performance after transients (see 4.4.6).
Spike voltage	Control unit shall return to normal operation without deterioration in performance after spikes on each phase (see 4.4.6).
Power interruption	Control unit shall return to normal operation without deterioration in performance when power is restored (see 4.4.6).
Insulation resistance	Control unit shall have a minimum insulation resistance of 10 megohms (see 4.4.6).
Leakage current	Control unit under normal operation shall not have a leakage current greater than 5 milliamperes (mA) (see 4.4.6).

3.4.8.2 Automatic supply dispenser. The supply dispenser shall automatically add supplies at the correct moment as directed by the formula control unit. The dispenser shall have not less than four supply compartments of sufficient capacity to hold supplies for one load of clothing. Three compartments shall hold dry supplies of detergent bleach (“two-shot”), starch, and sour blue. The supply in each compartment shall be completely water flushed into the cylinder when directed by the formula control unit.

3.4.9 Booster heater. Booster heater shall be located in the tub to remain fully submerged at inclination angles of 12 degrees at the lowest water level fill . Booster heater shall increase water temperature by 15*F per minute by indirect steam for type I and 10*F per minute by electric type II washer-extractor. The booster heater shall meet the minimal requirements when tested as specified in 4.4.5.

3.4.9.1 Indirect steam heater. Each type I machine shall be provided with a means of efficiently transferring the heat entering the machine at the steam connection to the wash and rinse solutions without the steam or its condensate coming into direct contact with the solution or wash load. A steam heat valve, strainer, and trap shall be furnished with the indirect heat steam piping system. Any component of the heat transfer equipment that contacts the wash or rinse solutions shall be stainless steel material; sheet or plating shall be as specified for the shell in 3.2.2; pipe shall be as specified in 3.2.5.1. Heat exchanger assembly shall be in accordance with ASME Boiler and Pressure Vessel Code, section VIII. The steam booster heater shall meet the minimum requirements of 3.4.9 when tested as specified in 4.4.5.

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3.4.9.2 Electric immeration heater. Tubular, electric, immersion heating elements on type II washer-extractor booster heaters shall be the screw plug type, sheathed In Incoloy (800 or 825) or Inconel (600 or 625) or equal. The spiral wound resistance wire shall be made of 80 percent nickel and 20 percent chromium suspended in a densely packed, magnesium oxide dielectric. Insulating material for formed parts, such as spacers, cores or bushings shall be high quality electrical grade ceramic of high insulation resistance and formulated for physical and thermal strength. Bending of the elements shall be done by the manufacturer of the immersion heating elements. The configuration of the immersion heating elements shall provide efficient heating of the tub water as specified in 3.4.9. The heating elements assembly shall be easily removed from the washer-extractor, The heating elements assembly shall be integrated with the temperature controls to form a system to meet the requirements specified in 3.4.8.1.2. The heating elements shall be 440 Vat, 60 Hz, 3-phase and shall be not greater than the power requirement specified in table IV.

TABLE IV. Power usage of heating element assembly

Normal capacity per load up to dry weight (pounds)	Maximum power (kW)
60	24
100	35
200	60

3.4.9.2.1 Terminal construction. The ends of the terminals shall be constructed to minimize the entrance of atmospheric moisture Terminal studs shall be of the threaded type or brazed lugs. Nuts and lockwashers for connections shall be furnished with the heating elements with threaded terminal studs.

3.4.9.2.2 Insulation resistance. The insulation resistance of the elements at any temperature from room to 1200°F shall be not less than the value computed by the following:

$$R - \frac{50-0.02T}{L}$$

Where:

R - Insulation resistance in megohms.
T - Temperature of element sheath in °F. When measuring insulation resistance (cold), T shall be room temperature.
L - Length of element from end to end in feet

For heating elements whose maximum operating temperature is in excess of 1000"F, the hot resistance shall be tested as specified in 4.4.7.2. Insulation resistance measurement tests specified in 4.4.7.1 shall be made at room temperature and at the maximum operating temperature. The cold and hot resistance shall be a minimum of 25 and 10 megohms respectively.

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3.4.9.2.3 Dielectric strength. Heating elements shall withstand a dielectric test of twice rated line voltage plus 1000 for a period of 1 minute. The voltage wave shall approximate a sine wave and the frequency shall be not less than 60 Hz.

3.4.10 Miscellaneous requirements.

3.4.10.1 Idendification and information plates. An identification and an information plate in accordance with MIL-P-15024 and MIL-P-15024/5 shall be securely affixed to each machine in a conspicuous place. The plates shall be made of commercial corrosion resisting steel, or brass. All q arkings shall be legible. Data marked on the identification plate shall include the following:

(a) Manufacturer's type or model number, serial number, contract or order number, and date of manufacture.
(b) Salient design features including size, speed, voltage, current, and capacity in pounds - dry weight per load.
(c) Name of machine and stock number (the stock number will be furnished to the contractor).
(d) The component identification number entry shall consist of the letters "CID", followed by a nine digit number, which will be furnished upon receipt of provisioning technical documentation (PTD). No equipment shall be shipped without the CID number for the washer-extractor, and its associated components shall also be recorded on the packing list.

3.4.10.2 Information plate. Data marked on the information plate shall be in 3/8-inch letters or larger and include the following, if of the three-pocket type (A similar plate shall be provided for the other sized machines).

LOADING PROCEDURE - 60-POUND MACHINE
AN EQUAL AMOUNT OF DRY LAUNDRY SHALL
BE PLACED IN EACH POCKET NOT TO EXCEED
20 POUNDS IN EACH POCKET

3.5 Workmanship. The materials for construction, fabrication, finish and operation shall be of the highest quality commensurate with the intended use. Tolerances shall be in accordance with good commercial standards. The interior and exterior of the machines shall be thoroughly cleaned and finished. Workmanship shall be equivalent to the best in the industry and shall conform to the following:

(a) Welds shall be cleaned and ground to present a uniform and smooth finish.
(b) Parts shall be tight and of rigid construction so that the machine does not leak or rattle during operation.
(c) Cylinder, with load and without, shall not visibly wobble during operation.
(d) Controls and door handle shall be located and hazard free to operator.

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(e) Surfaces shall be free of burrs and sharp edges.
(f) Safety guards shall be adequate and suitably located.
(g) Drive adjustment mechanism shall be suitable for the purpose intended.
(h) Bearings, gears, and shafts shall be properly constructed, aligned, and fitted for durable operation.
(i) Lubrication fittings shall be provided where necessary, shall be adequate for the purpose intended, and easily accessible.
(j) Fastenings shall be adequate and accomplish the purpose for which they are intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibiliy for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective q aterial, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.5).

4.3 First article inspection First article inspection shall consist of the examinations shown in table V and tests specified in 4.4.1 through 4,4.8 (see 6.3).

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TABLE V. Classification of defects.

Examination	Defect (requirement)	Classification	
		Major	Minor
Materials	Construction materials not as specified (see 3.2 through 3.2.7).	X	
Design, general	Configuration not as specified (see 3.3 through 3.3.10).	X	
Automatic programmer	Does not provide control as specified (see 3.4.8.1 through 3.4.8.1.12).	X	
Displays and signals	Displays and signals not as specified (see 3.4.8.1.3, 3.4.8.1.6, 3.4.8.1.4, 3.4.7.4.9, 3.4.7.4.7 and 3.4.8.1.10).	X	
Water temperature	Not as specified (see 3.4.8.1.3).	X	
Water level control	Does not provide for three levels (see 3.4.8.1.3).	X	
Door interlock and safety devices	Not provided, not as specified (see 3.4.7.4).	X	
Emergency stop button	Not provided, not as specified (see 3.4.7 4.9).	X	
Capacity, size, weight	Machine not constructed for specified dry weight capacity (see 3.3.1.7).	X	
	Machine exceeds dimensions and weight limits (see 3.3.1.7).	X	
Construction/ mechanical	Mounting holes not provided, not as specified (see 3.3.3).	X	
	Shell openings not as specified (see 3.4.1.1).	X	
	Drain valve not as specified (see 3.4.4 and 3.4.8.1.8).	X	
	Piping, valves, and fittings not as specified (see 3.2.5 through 3.2.5.5 and 3.4.5 through 3.4.5.4).	X	
	Overflow connection not as specified (see 3.4.5.3).	X	
	Cylinder bearings and seals not replaceable (see 3.3.6.3).	X	
	V-belts not properly sized or not adjustable (see 3.3.6.4).	X	
Sectionalization	Sectionalization not as specified, if required (see 3.3.10)	X	
Construction/ electrical	Motor not as specified (see 3.4.7.1).	X	
	Electrical connection boxes not dripproo enclosed or better (see 3.4.7.2).	X	
	Electrical terminal blocks not marked to correspond with electrical diagram (see 3.4.7.3).	X	
	Wires not properly marked to correspond with electrical diagram (see 3.4.7.3).	X	

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TABLE V. Clasification of defects - Continued.

Examination	Defect (requirement)	Classification	
		Major	Minor
Construction/ electrical	Immersion electric booster heater exceeds power requirements as specified (see 3.4.9.2).	X	
	Pin connector to microprocessor not as specified (see 3.4.8.1.2).	X	
	Booster heater not as specified (see 3.4.9 through 3.4.9.2).	X	
	Microprocessor control layout not as specified (see 3.4.8.1.10, through 3.4.8.1.10.3).	X	
Identification	Identification and information plates not of specified material (see 3 4.10 1)		X
	Marking illegible (see 3 4.10.1)	X	
	Identification and information plates not provided as specified (see 3.4.10.1 through 3.4.10.2.1).	X	
Safety features	Belts, shafts, pulleys not properly enclosed (see 3.4.7.4.6).	X	
	Door interlock safety device not provided or not as specified (see 3.3.1.4 and 3.4.7.4.4).	X	
Workmanship	Surfaces of cylinder not free of burrs and sharp edges (see 3.4.2).	X	
	Rust, toolmarks, pits, scratches, wrinkles present (see 3.5).		X
	Welds not buffed (see 3.5).		X
	Surfaces not free from burrs, sharp edges (see 3.5).		X

4.3.1 First article sample. Prior to the start of production, one machine of each configuration and size shall be made ready for tests and examinations specified in 4.4.1 through 4.4.8 (see 6.6).

4.4 First article tests.

4.4.1 Inclination. The machine shall be inclined at an angle of 12 degrees to each side of two vertical planes at right angles to each other and shall operate normally through the wash formula (see table VII) with a capacity load in each of the four positions (see 3.3.4). The overflow may be blanked off during this test.

4.4.2 Vibration test. Washer-extractor shall be tested in accordance with the requirements specified in MIL-STD-167-1 for type I vibration. Washer-extractor shall be operationally run during and after testing.

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4.4.3 Extract efficiency. The machine shall extract water to the extent that the water retention of material shall be not greater than 55 percent after 5 minutes of extraction. The water temperature of the bath immediately preceding each extraction shall be not greater than 135*F. The average of 10 runs shall be used to evaluate the performance of the machine under this requirement.

4.4.3.1 Load. A capacity load of cotton muslin sheets of 100 percent cotton fiber free of filler or sizing shall be used in the extraction efficiency tests. The sheets shall be washed through four complete washing cycles, using commercial laundry soap, builder, and bleach in accordance with the contractor's instructions , and then dried to less than 4 percent moisture content to determine dry weight prior to conducting the first extraction test.

4.4.4 Airborne noise. The washer-extractor shall be tested for airborne noise produced during operation in accordance with the procedures specified in MIL-STD-740-1. Measurements shall be for sound pressure level determination only. The washer-extractor shall be tested under full load conditions in both the washing and extraction modes of operation. The washer-extractor shall meet the requirements of 3.3.8.

4.4.5 Water heating. The water heating system shall heat the wash bath at a minimum of 15*F per minute for type I washer-extractors and 10"F per minute for type II washer-extractors (see 3.4.9). Test conditions and procedures shall be in accordance with the following:

(a) Conditions - Ambient temperature - 65 to 85*F.
Inlet water temperature - 50 to 75*F.
Steam, type I (saturated) - 90 to 100 psi.
Electric, type II - 440 Vat, 60 Hz, 3-phase, maximum power (see table IV).
Water level - 4 inches (determined by measurement, not sight glass observation).

(b) Setup procedure - A thermocouple shall be installed in the lowest portion of the shell and be extended not less than 3 inches into the shell. A digital temperature readout indicator with plus or minus 1*F accuracy shall be connected to the thermocouple.

(c) Test procedure - Fill machine cylinder to 8 ± 1/4 inches water level with cold water. Activate steam or electric heating system and energize wash motor. When water temperature indicator reads 100"F start timing device (device shall have graduations of not greater than 1 second) . Stop timing device when temperature indicator reads 150°F Record elapsed time De-energize wash motor and heating system Recheck water level, if not within the limits of 8 ± 1/4 inches, disregard test. Open drain valve for 2 minutes or 1 minute longer than the time required for complete drainage of shell, whichever is longer. Repeat above procedure until five runs are complete.

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(d) Test results - Rate of rise shall be determined by adding the elapsed times for the 100 to 150°F temperature rise for the five runs and divided into 250. A temperature rise less than 15*F per minute for type I washer-extractors and 10"F per minute for type II washer-extractors constitutes failure of test.

4.4.6 Test for automatic formula control unit. The automatic formula control units shall be tested and meet the requirements specified in 4.4.6.1 through 4.4.6.11.

4.4.6.1 Low temperature test. Unit shall be tested in accordance with q ethod 502, procedures I and II, of MIL-STD-810. Unit shall be tested at a constant storage temperature of minus 4°F for 24 hours, and a constant operating temperature of 32°F for 24 hours.

4.4.6.2 High temperature test. Unit shall be tested in accordance with method 501, procedures I and II, of MIL-STD-810 Unit shall be tested at a constant storage temperature of 160" for 24 hours, and a constant operating temperature of 122°F for 24 hours.

4.4.6.3 Humidity. Unit shall be tested in accordance with method 507, procedure 1, of MIL-STD-810. Unit shall be tested at a relative humidity of 95 percent or higher at temperature range of 86 to 122°F.

4.4.6.4 Steady state voltage and frequency. The equipment shall be operated for not less than 15 minutes in each of the conditions A thru E of table VI. Performance readings shall be measured and recorded for each condition. Failure in any performance parameter shall be cause for rejection.

TABLE VI. Steady state voltage and frequency.

Condition	Rated voltage (± 10%)			Frequency (60 Hz ± 5%)		
	Lower limit	Normal	Upper limit	Lower limit	Normal	Upper limit
A (reference condition)		X			X	
B	X			X		
C			X	X		
D			X			X
E	X					X

4.4.6.5 Transient voltage.

4.4.6.5.1 Upper limit. With the equipment operating in the upper limit of steady state voltage, a transient voltage of plus 18 percent of normal voltage recovering to the steady state voltage within 2 seconds shall be superimposed. The equipment shall be capable of normal operation following the transient.

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4.4.6.5.2 Lower limit. With the equipment operating on the lower limit of steady state voltage, transient voltage of minus 18 percent of normal voltage recovering to the steady state voltage within 2 seconds shall be superimposed. The equipment shall be capable of normal operation following the transient.

4.4.6.5.3 Transient (700 V). In addition to the tests of 4.4.6.5.1 and 4.4.6.5.2, equipment containing transistors or semi-conductor devices shall be subjected to an input supply line transient of 700 V amplitude and 2 milliseconds duration while the equipment is operating at normal supply line voltage and frequency. The equipment shall be capable of normal operation immediately following the transient.

4.4.6.6 Tansient freguency.

4.4.6.6.1 Upper limit. With the equipment operating at 5 percent above the normal frequency, the frequency shall be increased by an additional 3 percent, recovering to the steady state frequency (plus or minus 5 percent of normal) within 2 seconds. The equipment shall be capable of normal operation following the transient.

4.4.6.6.2 Lower limit. With the equipment operating at 5 percent below the normal frequency, the frequency shall be decreased by an additional 3 percent, recovering to the steady state frequency (plus or minus 5 percent of normal) within 2 seconds. The equipment shall be capable of normal operation following the transient.

4.4.6.7 Spike voltage test. In addition to the tests of 4.4.6.6.1 and 4.4.6.6.2, the equipment shall be subjected to an input supply line voltage spike of 2500 V positive peak amplitude; the wave shape shall correspond with that of the figure for the “spike voltage (short time transient) wave shape”, section 300 of MIL-STD-1399. This spike shall be impressed at normal supply voltage and frequency while the equipment is in operation. The equipment shall be capable of normal operation immediately following the test.

4.4.6.8 Power Interruption. With the equipment operating within the steady state tolerances of voltage and frequency, the external power supply shall be suddenly interrupted and after an interval between 3 and 4 seconds, the power supply, within the steady state tolerance performance degradation and to include any recycling time, the power shall be interrupted repeated for a total of four times. During and as a result of these tests, no damage to the equipment shall be incurred, and any noted effects of power interruption or reapplication shall fall within the performance tolerances of the individual equipment specification. Where an equipment has greater than one significant operating mode, the power interruption tests shall be performed in each of the modes.

4.4.6.9 Insulation resistance. The insulation resistance of electrical circuits shall be not less than 10 megohms at 500 volts direct current (Vdc) at a temperature of 23 ± 1 4 degrees Celsius (°C) and at a relative humidity of 50 ± 50 percent. All synchros shall be disconnected during this test.

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4.4.6.10 leakage current. The leakage current between the power line and equipment enclosure shall be measured through a 1500 ohm resistive load at normal power line voltage. Leakage current of the equipment shall be not greater than 5 mA to ground.

4.4.6.11 Enclosure. Unit shall be tested in accordance with MIL-STD-108, "drip test". Maximum angle of inclination shall be 45 degrees.

4.4.7 Dielectric strength test. The electric heating elements shall withstand, for a period of 1 minute, a dielectric test voltage of twice the rated line voltage plus 1000 volts applied between the terminal and the sheath as specified in 3.4.9.2.3. The test shall be conducted with the heating element at normal ambient temperature. If shorting occurs the element has failed the test.

4.4.7.1 Insulation resistance, cold. The insulation resistance shall be measured with 500 Vdc at normal ambient temperature to determine conformance with 3.4.9.2.2.

4 4.7.2 Insulation resistance, hot. The Insulation resistance shall be measured with a minimum of 500 Vdc within 5 seconds after operation of the element at maximum operating temperature of the element to determine conformance with 3.4.9.2.2. To prevent the element from being cooled before the test, the ambient fluid shall not be circulated after the power is removed.

4.4.8 Electromagnetic interference tests. MIL-STD-461, class 4A, tests CE01, CE03, and RE02 shall be performed on each machine, graphical limiting emissions for surface ships shall be in accordance with MIL-STD-461, Test setups shall be in accordance with MIL-STD-462 (see 3 4 7.3.1).

4.5 Quality conformance inspection Quality conformance inspection shall consist of the examination and tests specified in 4.6 and 4.7 (see 6.3).

4.5.1 Sampling for quality conformance.

4.5.1.1 Lot. For the purpose of quality conformance examination and test sampling, a lot is defined as all the machines of the same design, type, style, class, and size, produced in one facility, using the same production processes and materials, and being offered for delivery at one time.

4.5.1.2 Sampling for quality conformance examinations and tests. As a minimum, the contractor shall randomly select a sample quantity from each lot of completed machines in accordance with tables VII and VIII and examine and test shall be in accordance with 4.6 and 4.7. If one or more defects are found in any sample, the entire lot represented by the sample shall be rejected. If a lot is rejected, the contractor has the option of screening 100 per cent of the lot for defective characteristic or providing a new lot which shall be examined and tested in accordance with the sampling plans contained herein. The contractor shall maintain for a period of three years after contract completion all records of inspections, tests, and any resulting rejections.

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TABLE VII. Sampling for quality conformance examination.

Lot size	Sample size	
	Major defects	Minor defects
2 to 25	5	3
26 to 50	5	5
51 to 90	7	6
91 to 150	11	7
151 to 280	13	10
281 to 500	16	11
501 to 1200	19	15
1201 to 3200	23	18

TABLE VIII Sampling for quality conformance tests.

Lot size	Sample size
2 to 8	All
9 to 150	13
151 to 280	20
281 to 500	29
501 to 1200	34
1201 to 3200	42

4.6 Examination. Sample machines shall be examined to verify conformance to the requirements of this specification. Intermediate products or assem- blies which cannot be checked or measured after final assembly shall be examined at various stages of manufacture for conformance to table V. Examinations shall be made at appropriate times for defects in construction, workmanship, dimensions, and materials. Any deviation from the specified requirements shall be classified as a defect. The inspection shall be level II (see 6.9.1).

4.7 Quality conformance tests.

4.7.1 Insulation. The insulation resistance of each motor, controller, and connecting cable shall be measured by a 500 Vdc insulation resistance tester. Insulation resistance of each component shall be not less than 10 megohms.

4.7.2 Inching. With the shell door open, determine that the cylinder is not capable of rotating except during inching. With the cylinder rotating, determine that the shell door cannot be opened.

4.7.3 Power input. Motors of each machine shall not draw in excess of rated current after accelerating to full speed.

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4.7.4 Braking. It shall be determined that a normal full rated load can be brought to rest from the extracting speed within 60 seconds without deleteriously affecting the brake (see 3.4.3). The temperature rise of the brake lining shall be not greater than 300°F above the initial cold temperature of the brake lining as a result of three stops from peak speeds at 2 minute intervals during a loaded extraction cycle. The machine shall not extract when the brake interlock switch is applied.

4.7.5 Operation. The machine shall be operated for not less than three cycles using the formula specified in table IX. During this test it shall be verified that the valves, controls, switches, interlocks, gauges, timers, water tempera- ture, water levels, and automatic supply dispensers, and maximum water consump - tion (see 3.3.2) are functioning properly and that there are no water leaks in the shell, piping, and valves.

TABLE IX. Operational test formula

Step	Operation	Time (minutes)	Water temperature	Water level Type I/ Type II (inches)	Supplies/100-pound washer/extractor basis
1	Break suds	10	160 ± 5°F	4/8 ± 1	16 oz. detergent/ oxygen bleach
2	Drain	1	---	---	---
3	Flush/suds	6	160 ± 5°F	4/8 ± 1	
4	Drain	1	---	---	---
5	Spin	1	---	---	---
6	Rinse	3	160 ± 5°F	4/8 ± 1	---
7	Drain	1	---	---	---
8	Rinse	3	160 ± 5°F	4/8 ± 1	---
9	Drain	1	---	---	---
10	Sour	4	130 ± 5°F	4/8 ± 1	2 oz. sour/blue
11	Drain	1	---	---	---
12	Final spin	4	---	---	---

4.7.6 Hydrostatic testing. team booster and steam lines shall be hydro- statically tested to a pressure of 150 psi for a minimum of 30 minutes. In both cases no leaks in either system shall be permitted.

4.8 Inspection of packaging. Sample packages and packs, and the inspec- tion of the preservation, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6,8.)

5.1 Preservation, packing and marking. Washer-extractor shall be preserved level A or commercial, packed level A, B, or commercial, as specified (see 6.2), and marked for each level of protection in accordance with MIL-L-3153. Washer-extractors shall be shipped fully assembled, unless disassembly is necessary to protect certain components during transit, or disassembly results in greater than a 25 percent savings in transportation costs to the Government or to the contractor. In addition, for Navy acquisitions, the following applies:

(a) Navy fire-retardant requirements.

(1) Lumber and plywood. When specified (see 6.2), all lumber and plywood Including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.
Level C - Type I - non-weather resistant.
Category 1 - general use.

(2) Fiberboard. Unless otherwise specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, and cleated fiberboard boxes including interior packaging forms shall meet the flame spread index and specific optic density requirements of PPP-F-320 and amendments thereto.

5.2 Technical manuals. Two copies of the current operation/maintenance technical manual(s) shall be packaged with each system or equipment. Technical manual packaging shall be as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Commercial laundry washer-extractor machines acquired under this specification are intended for shipboard use. Therefore, commercial machines offered to the Government may require alteration to meet specification requirements, for example, saltwater input, steam injection, proper materials, wiring, electromagnetic emission requirement, vibration requirements, electric booster heater.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, style, capacity, size and weight of machine required (see 1.2 and 3.3.1.7) and type of controller mounting (see 3.4.8.1).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) When first article inspection is required (see 3.1).
(e) Level of preservation, level of packing, and packaging options required (see 5.1).
(f) When fire retardant treated lumber and plywood is required (see 5.1(a)(1)).
(g) When fire retardant fiberboard is not required (see 5.1(a)(2)).
(h) Technical manual packaging requirements (see 5.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.3	DI-NDTI-80603	Test procedure	----
4.5	DI-MISC-80678	Certification/data report	10.3.1 does not apply
4.5	DI-NDTI-80809	Test/inspection reports	----
6.6	DI-V-7002	Provisioning parts list	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) q ust be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

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6.4.1 Technical manuals development. Technical manuals (including technical repair standards (TRS)) should be procured under separate CLINs or subCLINs. Technical manual products (such as manuscript copy, preliminary copy, and reproducible copy) should be developed and delivered as specified in the Technical Manual Contract Requirements (TMCR) exhibit(s) to the contract. The TMCR is an exhibit or attachment to a contract specifying technical manual product development and delivery requirements, and/or management data development requirements. CDRLs and DID's should not be used for technical manual product procurements. Technical manual management data (such as plans and reports) should be procured under a separate data CLIN, but delivery should continue to be specified on CDRLs and DID's. Technical manual quality assurance management data should be prepared and delivered as specified in the TMCR and contractor order

6.5 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit of each configuration and size. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.6.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.7 National Stock number. The national stock number shown in the acquisition document is for reference only. Contractors will be advised of the applicable national stock number to be used on identification plates.

6.8 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.9 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

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6.10 Part of identifying number (PIN). The PIN to be used for lubricating oil acquired to this specification is created as follows:

M19044 - X X X
- Style a = 1, b = 2
- Class 1 or 2
- Type I = 1, II = 2

6.11 Subject term (key word) listing.

Automatic control
Electric booster
Electrical mechanical
Heater
Microprocessor
Steam booster

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 3510-N136)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form

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I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-W-19044E(SH)	2 DOCUMENT DATE (YYMMDD) 1993 September 18

3. DOCUMENT TITLE
WASHER-EXTRACTOR, LAUNDRY, 35 TO 200 POUND CAPACITY, MODIFIED COMMERCIAL (SHIPBOARD)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b ORGANIZATION	
c. ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Mr. Joseph Bowen, NAVSSES, Code 972D	(215) 897-7925	443-7925
c ADDRESS (Include Zip Code) Commanding Officer, Naval Ship Systems Engineering Station, Carderock Div. NAVSURFWARCEN, Philadelphia, PA 19112-5083	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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